

RIZHAO, China

WMO: 549450

Lat: 35.4681N

Lon: 119.5550E

Elev: 121

StdP: 14.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	22.5	-7.0	3.8	28.2	-2.9	4.7	27.8	22.6	29.0	20.3	31.6	6.7	320	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	89.1	77.7	86.6	77.2	84.6	76.6	81.7	86.2	80.6	84.3	79.5	82.8	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	159.1	84.5	79.4	153.4	83.4	78.5	148.4	82.3	45.7	86.4	44.5	84.5	43.3	82.8	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.7	16.6	14.3	15.2	94.1	3.6	3.5	12.7	96.6	10.6	98.7	8.6	100.7	6.0	103.2		
			DB	WB	12.0	83.4	3.3	1.1	9.7	84.2	7.7	84.9	5.9	85.5	3.5	86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	33.0	36.4	45.0	55.3	65.2	71.7	78.4	78.9	72.1	62.3	49.1	37.4	(6)
(7)		DBStd	16.91	5.60	5.69	6.42	6.12	5.75	4.46	3.97	3.71	4.16	5.51	7.22	6.17	(7)
(8)		HDD50	1605	527	381	185	16	0	0	0	0	2	102	392		(8)
(9)		HDD65	4236	992	801	622	298	69	3	0	3	114	478	856		(9)
(10)		CDD50	4223	0	0	29	176	470	650	880	895	664	384	74	1	(10)
(11)		CDD65	1379	0	0	0	8	74	203	415	430	217	32	0	0	(11)
(12)		CDH74	9672	0	0	1	61	431	895	3427	3854	956	47	0	0	(12)
(13)		CDH80	2257	0	0	0	17	109	216	881	921	111	2	0	0	(13)
(14)	Wind	WSAvg	6.4	6.7	6.7	6.9	7.1	6.4	5.7	5.5	6.0	6.2	6.2	6.7	6.8	(14)
(15)	Precipitation	PrecAvg	31.5	0.4	0.8	0.9	1.6	3.2	3.5	7.4	6.7	3.5	1.5	1.3	0.6	(15)
(16)		PrecMax	48.1	1.9	2.4	2.8	4.3	9.7	8.1	17.1	14.4	12.0	6.5	8.3	2.2	(16)
(17)		PrecMin	20.3	0.0	0.1	0.0	0.2	0.1	0.2	2.8	1.7	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	8.0	0.5	0.6	0.7	1.0	2.1	2.2	3.3	3.4	3.1	1.7	1.9	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	55.4	70.5	81.0	89.1	90.1	93.1	91.7	86.7	78.6	68.2	56.1	(19)
(20)		0.4%	MCWB	41.1	44.5	54.1	61.8	64.7	69.7	81.5	80.0	73.6	64.4	56.6	49.8	(20)
(21)		2%	DB	45.7	50.3	63.9	73.2	82.5	85.5	89.2	88.1	82.7	74.8	64.4	52.2	(21)
(22)		2%	MCWB	39.9	41.4	50.7	58.5	63.5	70.7	80.1	79.6	72.2	63.3	56.5	46.9	(22)
(23)		5%	DB	43.4	47.2	58.6	68.8	78.7	81.9	86.6	86.1	80.6	72.6	62.2	49.3	(23)
(24)		5%	MCWB	38.3	40.4	48.0	56.1	62.3	70.8	79.0	78.6	71.4	63.4	55.4	43.8	(24)
(25)		10%	DB	41.4	44.8	54.3	64.9	75.0	78.9	84.5	84.4	78.9	70.7	60.0	46.7	(25)
(26)		10%	MCWB	36.6	39.3	45.3	53.9	62.0	69.5	78.4	77.9	70.2	62.6	53.2	41.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.5	47.1	56.9	65.3	69.9	78.5	83.2	83.2	79.6	69.8	62.1	52.5	(27)
(28)		0.4%	MCDWB	47.3	51.6	67.2	75.6	79.1	84.2	89.5	87.8	82.9	73.1	63.7	53.8	(28)
(29)		2%	WB	41.9	44.3	52.7	60.8	67.4	75.5	81.7	81.9	76.4	67.7	59.5	48.1	(29)
(30)		2%	MCDWB	44.3	47.7	60.5	69.9	76.0	80.4	86.9	85.9	79.8	71.8	62.9	50.9	(30)
(31)		5%	WB	39.6	42.2	49.6	58.1	65.8	73.4	80.5	80.9	74.4	66.1	56.9	45.0	(31)
(32)		5%	MCDWB	42.5	45.3	56.0	65.2	73.7	77.7	84.9	84.4	78.1	70.6	60.9	48.5	(32)
(33)		10%	WB	37.2	40.4	47.1	56.1	64.2	71.7	79.4	80.0	72.7	64.2	53.9	41.9	(33)
(34)		10%	MCDWB	40.8	43.6	52.7	61.7	71.1	75.4	83.1	83.1	76.9	69.5	58.8	45.9	(34)
(35)	Mean Daily Temperature Range		MDBR	11.4	11.1	12.1	12.5	12.3	9.4	8.1	8.4	10.5	12.0	12.1	11.5	(35)
(36)		5% DB	MCDBR	13.2	14.2	18.9	19.2	19.7	15.5	11.7	10.3	12.3	13.3	13.5	13.4	(36)
(37)		5% DB	MCWBR	9.7	9.6	10.5	10.3	8.7	7.0	5.4	5.1	6.1	7.8	9.7	10.2	(37)
(38)		5% WB	MCDBR	11.6	11.6	16.4	16.0	15.3	11.0	9.6	8.1	9.1	10.5	11.6	11.5	(38)
(39)		5% WB	MCWBR	9.8	9.1	10.3	9.7	8.3	5.9	5.2	4.6	5.9	8.0	9.9	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.523	0.585	0.624	0.634	0.651	0.757	0.733	0.679	0.602	0.627	0.589	0.526	(40)	
(41)		taud	1.753	1.662	1.614	1.642	1.647	1.501	1.601	1.692	1.791	1.660	1.684	1.763	(41)	
(42)		Ebn at Noon	197	202	209	218	217	194	198	205	213	188	176	185	(42)	
(43)		Edh at Noon	56	69	78	80	80	93	83	75	64	68	60	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	793	980	1387	1623	1751	1650	1474	1455	1324	1119	830	716	(44)	
(45)		RadStd	87	126	110	115	137	130	158	168	117	121	103	67	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A

Nomenclature: See separate page